RAHUL SHARMA

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Summary — AI/ML Engineer with 2+ years of experience and a strong foundation in AI, machine learning, and software development. Experienced in fine-tuning vision-language models, building retrieval and agentic systems, and on-device speech recognition. Proven track record across research and production, currently working as an SDE-2 AI/ML Engineer at ZysecAI.

Skills

Programming Languages: Python, Rust, C/C++, SQL
Libraries & Frameworks: PyTorch, Transformers, TensorFlow, Scikit-Learn, NumPy, Pandas, FastAPI, LangGraph, DSPy, MLflow
LLM Inference & Serving: vLLM, SGLang, llama.cpp, MLX, Core ML, ONNX Runtime / sherpa-onnx
Tools & Platforms: AWS, Azure, GCP, Docker, Kubernetes, Linux, Git/GitHub, Hugging Face Hub, Jupyter
Databases: Milvus, Neo4j, FalkorDB, ClickHouse, DuckDB, MongoDB, MinIO, SQLite
Machine Learning: Vision-Language Models, OCR, Speech Recognition (ASR), LLM Fine-tuning, RAG, Agentic Systems (MCP), Model Quantization & On-device Inference, Classical ML (Classification, Clustering), Optimization & Scheduling, Models & Evals, Benchmarking

Experience

ZysecAI **Oct 2024 – Present**
SDE-2 AI/ML Engineer *Jan 2026 – Present*

- Designed **OCR-enabled ETL pipelines** for unstructured data, processing corpora from **hundreds of GBs to TBs**.
- Fine-tuned **vision-language models** for OCR and document understanding (**IBM Granite, SmolVLM, Qwen-VL**).
- Built a state-of-the-art **agentic graph RAG** system scoring ∼**92% on HotpotQA**, and retrieval pipelines that **reduced hallucinations by 95%** in production question answering.
- Built an **agentic cognitive memory system** for agents, scoring **over 96% on LongMemEval and LoCoMo** for long-term recall.
- Built scalable **agentic systems** that integrate **MCP servers** with dynamic workflows for incident triage, compliance review, security analysis and LEA workflows.
- Built **evaluation** for models and agentic systems.
- Authored research papers on agentic systems, including **HyperMind**.
- Built ML on **large-scale structured transport data** (millions of scheduling and runtime records): data filtering, cleaning and augmentation, **classification**, and **K-means clustering** for business analytics.
- Built **agentic optimizer workflows** for transport scheduling and planning, using **scheduling and optimization algorithms** for routing problems such as the **travelling salesman problem**.

SDE-1 AI/ML Engineer *Jan 2025 – Dec 2025*

- Architected **end-to-end AI pipelines** for enterprise integration, including data-processing workflows and advanced retrieval for AI agents at scale.
- Led **fine-tuning** across **NER**, **small language models** and **multimodal vision-language models**.
- Built **evaluation pipelines** to benchmark model and end-to-end pipeline performance.

Associate Generative AI Engineer *Oct 2024 – Dec 2024*

- Developed **Retrieval-Augmented Generation (RAG)** models to improve information retrieval and response accuracy for cybersecurity applications.
- Utilized **LangGraph** to build **agentic applications** that support real-time decision-making tailored for cybersecurity needs.
- Optimized **large language models (LLMs)** to manage memory constraints effectively, enhancing scalability and performance for complex, context-heavy tasks.
- Integrated **Neo4j** for **graph retrieval** to improve context handling in LLMs, allowing for enhanced understanding and response in multi-turn dialogues.

HacktivSpace **Sept 2024 – Mar 2025**
Machine Learning Engineer (Part-time, Community Contributor)

- Worked on **Retrieval-Augmented Generation (RAG)** for context retrieval in EdTech applications, utilizing **Milvus** and **Neo4j** to manage and retrieve data effectively.
- Developed **conversational memory** in LLM applications to maintain context across sessions, improving engagement and continuity in AI interactions.
- Used **LangGraph** to create **AI agentic workflows** for various tasks, supporting automation and enhanced responsiveness.

HKBK College of Engineering **Aug 2023 – May 2024**
Undergraduate Researcher

- Developed a model to analyze and respond to questions on **biomedical images** with **over 70% accuracy**.
- Conducted research on fine-tuning large language models like **LLaMA** for medical question-answering.
- Designed an **R-CNN**-based model to extract templates and information from documents for fraud analysis.

Varcons Technologies **Aug 2023 – Sep 2023**
Machine Learning Intern

- Built a **sentiment-analysis** model on financial text and used its signals as features for stock price movement prediction.
- Conducted research on various machine learning topics in computer vision and language processing systems.

Projects

Vesper: On-device Voice-to-Text for macOS (Rust) **link**

- Built an **open-source desktop app** that transcribes dictation and audio files with **local models**, with optional text cleanup by a **local LLM**. No account or cloud API is needed.
- Ported speech models to **Apple Silicon** backends (Parakeet TDT 0.6B v3 for **MLX**, **Core ML** int8 on the Neural Engine and 4-bit **sherpa-onnx**; Moonshine) and published them on **Hugging Face**.
- **Benchmarked ASR models and small language models (SLMs)** for transcription and text cleanup on an open **multilingual ASR benchmark** dataset (English, Hindi, German, French, Spanish) published on Hugging Face.

IndicEAR: Speech Recognition for 11 Indian Languages + English **link**

- Turned the frozen audio encoder of **EmbeddingGemma 2** into a **speech recognizer** for code-switched speech, training only **11.8M parameters** of residual adapters, script-factored **CTC** heads and language-ID heads.
- Built a **custom streaming training pipeline** with **dynamic batching** (duration buckets, frame budget autotuned to VRAM, OOM backoff), **exact resume** (stream position, shuffle buffer and RNG saved in checkpoints), **bf16** and **on-GPU audio augmentation**, keeping the GPU **93–99% busy**.
- Trained on **8,356 hours of audio** streamed from the Hugging Face Hub in about **6 GPU-hours** (about 1,390 hours of audio per GPU-hour).
- Released the model, a **pip package** and an **open benchmark** (Vesper Indic Bench) scored against IndicConformer and SraVaani on identical clips.

Medi-Care: VQA for Medical Imaging (IJETS 2024 publication) **link**

- Developed a multimodal model for analyzing and responding to medical images using the **SLAKE** (radiology) and **PathVQA** (pathology) datasets.
- Fine-tuned **LLaMA** models for question answering, achieving **over 75%** accuracy across SLAKE and PathVQA.
- Incorporated **RAG** techniques to enhance model accuracy and credibility for knowledge-intensive tasks.

Education

Visvesvaraya Technological University **Bengaluru, India**
HKBK College of Engineering
Bachelor of Engineering(B.E/B.Tech) in Computer Science and Engineering *Sep 2020 – June 2024*
CGPA: 8.6/10

Publications

Beyond Imagery: AI-Enhanced Diagnostic Assistant for Cancer and Tumor Diagnosis using Radiology Imaging **link**
Authors: Dr. Nandha Gopal S M, Rahul Sharma, Nithin M, Prajwal B R, Prashanth Kalgonda *Mar 2024*
International Journal On Engineering Technology and Sciences (IJETS) *Code: Medi-Care*